



Blackout Math

Turjo and Urjo are two exceptionally good brothers. One lazy Friday afternoon, the power went out, leaving them completely bored with nothing to do. To pass the time, Urjo decided to invent a math game to challenge Turjo's calculating speed.

Urjo grabs a piece of paper and writes down two starting numbers, A and B . He then tells Turjo that he is generating a special sequence. However, since standard addition is way too easy for them, Urjo uses the Bitwise XOR operator ($\oplus$) to calculate the next numbers.

Let F be a sequence. The sequence is defined by Urjo as follows:

- $F_1 = A$
- $F_2 = B$
- $F_i = F_{i-1} \oplus F_{i-2}$ for all $i \geq 3$

The rules of the game are simple: Urjo will suddenly shout a number N . In order to win the game and prove he is the superior brother, Turjo must instantly shout back the exact value of the N -th term in the sequence defined as F_N .

He needs your help to write a lightning-fast program so he can beat his brother and win the game!

Input Format

Input is given in the following format:

- line 1: $N \ A \ B$

Output Format

- line 1: A single integer denoting the exact value of the F_N .

Constraints

- $1 \leq N \leq 10^{18}$
- $0 \leq A, B \leq 10^{18}$

Subtasks

Subtask	Score	Additional constraints
1	5	$N \leq 3$
2	7	$A = 1$ and $B = 1$ and $N \leq 10$
3	8	$N \leq 20$
4	10	$N \leq 1000$
5	30	$N \leq 10^5$
6	40	No additional constraints

Examples

Example 1

3 6 7

The correct output is

1

Example 2

5 9 11

The correct output is

11

Explanation

Example 1: $N = 3, A = 6, B = 7$

- $F_1 = 6$
- $F_2 = 7$
- $F_3 = F_2 \oplus F_1 = 7 \oplus 6 = 1$

Example 2: $N = 5, A = 9, B = 11$

- $F_1 = 9$

- $F_2 = 11$
- $F_3 = F_2 \oplus F_1 = 11 \oplus 9 = 2$
- $F_4 = F_3 \oplus F_2 = 2 \oplus 11 = 9$
- $F_5 = F_4 \oplus F_3 = 9 \oplus 2 = 11$

Note

Bitwise XOR ($\oplus$) compares two numbers in their binary representation. It evaluates each pair of bits according to the following truth table:

Bit A	Bit B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

For example, $6 \oplus 7 = 1$:

- 6 in binary is 00110
- 7 in binary is 00111
- Comparing them bit-by-bit yields 00001, which is 1 in decimal.